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Fig. 1 illustrates the intricate interaction of the thermal degradation processes at the electrolyte and the SEI. The reactions are strongly connected *via* the participating species, leading to a complex interconnected reaction network, including a reaction cycle involving HF and H_2O.

2.2 Mathematical model

This section presents the modelling approach, the modelled system and the underlying assumptions. Our previous publication[4] provides a more detailed description of the basic model.

2.2.1 Model overview. The modelled thermal runaway scenario is that of a Li-ion battery subjected to an ARC test: here, the battery is heated step-wise, and if the battery is not heating up on its own during a resting phase without heating due to degradation reactions (self-heating), a further heating step is conducted. Temperature evolution is recorded. For more details, see the Parameterisation section. The reference experiment[33] was conducted at open circuit voltage OCV = 4.15 V, corresponding to a state-of-charge of the battery of 100%. The battery was closed, *i.e.* no reactants could enter or leave the cell. Thus, the component balance in our model, eqn (1) in Table 2, accounts just for the reactions in Table 1, with no transport or further reactions with cell-external components. Reaction kinetics are based on an Arrhenius approach for almost all reactions, see eqn (2). Exceptions are the SEI formation reactions, *i.e.*, OSP, ISP, and LSP, where the active material LiC_6 reacts with electrolyte components. These reactions are known to slow down with increasing SEI thickness, as seen in eqn (3). Besides the classical Arrhenius type temperature dependence we therefore introduce a dependency of SEI forming reaction kinetics on the SEI thickness. This dependency should be understood as the capability of the SEI to form an electron-insulating layer. This capability is reduced by its decomposition. However, crack formation induced by particle swelling[38–40] and SEI dissolution[41] could also lead to new SEI components. As these effects are not directly modelled, they are implicitly included in this formulation. Furthermore, the kinetics of the conductive salt decomposition (CSD) includes α, which is the dissociation constant of $LiPF_6$. Additionally CSD and the subsequent decomposition of PF_5 (PFD) are modelled as reversible reactions, see eqn (4) and (5). For both, the rate constant for the backward reaction is calculated based on the equilibrium constant, which in turn is computed from the Gibbs free energy of the respective reaction as seen in eqn (7)–(9).[42] For the simulation of ARC measurements, we follow

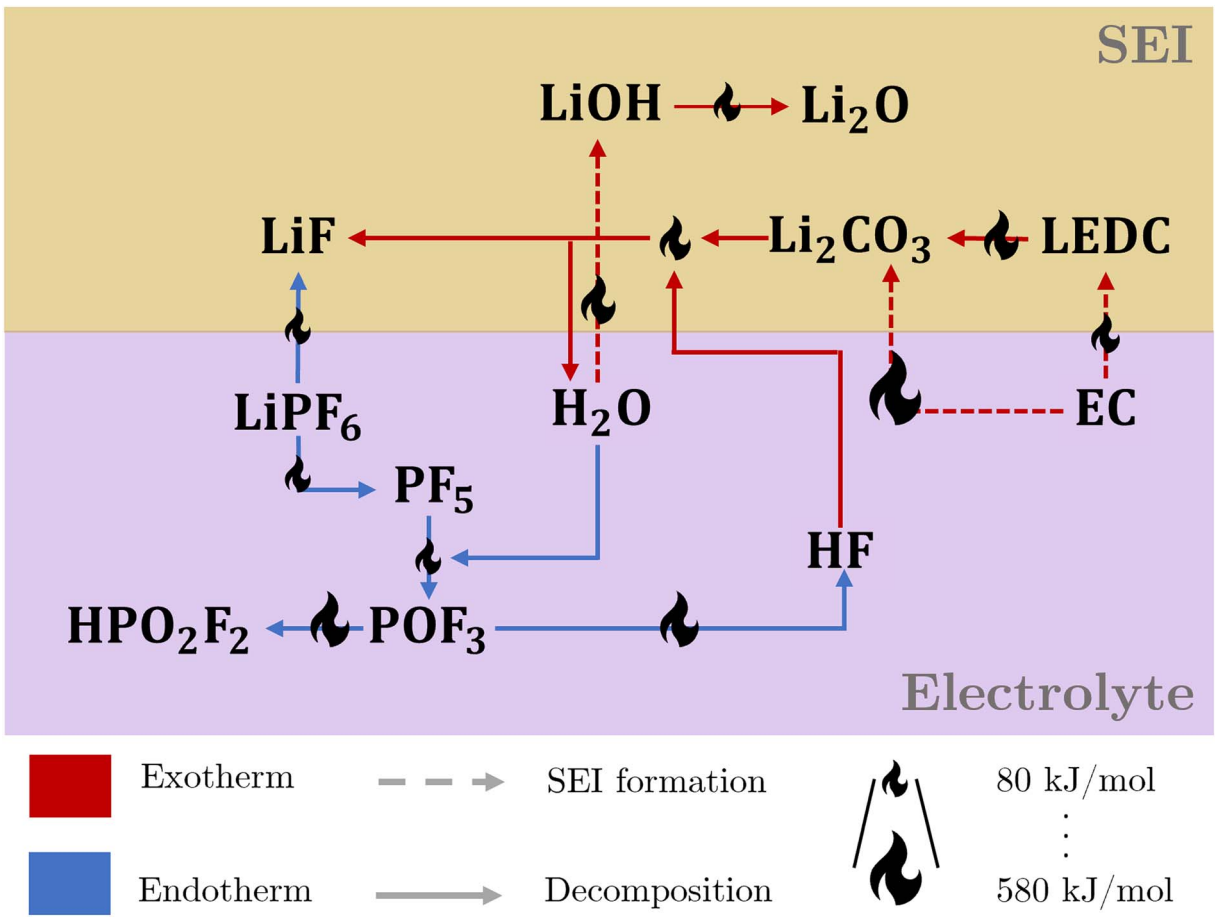

Fig. 1 Reaction network for thermal composition of the SEI and electrolyte, covering the interaction between salt, solvent, and SEI. Red and blue arrows indicate exo- and endothermic reactions, respectively. The size of the flame icon on the arrows scales with the corresponding amount of consumed or released reaction heat.

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Table 2 Model equations to simulate the ARC test. They have been adopted from Baakes *et al.*[4] Ω is denoted as the set of species (SP), reactions (RE), a subset of species i participating in reaction j (U), gaseous species (G), liquid species (L), and solid species (S)

Description	Equation	No.
Species balance	$\frac{\mathrm{d}n_i}{\mathrm{d}t} = \sum_{j\in\Omega_{\mathrm{RE}}} \nu_{j,i} r_j,\ \forall i \in \Omega_{\mathrm{SP}}$	(1)
Reaction kinetics	$r_j = k_{\mathrm{f},j} \prod_{u\in\Omega_{\mathrm{U_f}}} a_u^{\nu_{u,j}},\ \forall j \in \Omega_{\mathrm{RE}}/\{\mathrm{OSP,\ ISP,\ LSP,\ CSD,\ PFD}\}$	(2)
	$r_j = k_{\mathrm{f},j} \frac{1}{d_{\mathrm{SEI}}} \prod_{u\in\Omega_{\mathrm{U_f}}} a_u^{\nu_{u,j}},\ \forall j \in \{\mathrm{OSP,\ ISP,\ LSP}\}$	(3)
	$r_j = k_{\mathrm{f},j}(1-\alpha) \prod_{u\in\Omega_{\mathrm{U_f}}} a_u^{\nu_{u,j}} - k_{\mathrm{b},j} \prod_{u\in\Omega_{\mathrm{U_b}}} a_u^{\nu_{u,j}},\ \forall j \in \{\mathrm{CSD}\}$	(4)
	$r_j = k_{\mathrm{f},j} \prod_{u\in\Omega_{\mathrm{U_f}}} a_u^{\nu_{u,j}} - k_{\mathrm{b},j} \prod_{u\in\Omega_{\mathrm{U_b}}} a_u^{\nu_{u,j}},\ \forall j \in \{\mathrm{PFD}\}$	(5)
	$k_{\mathrm{f},j} = k_{0,j} \exp\left(-\frac{E_{\mathrm{A},j}}{\mathrm{R}T}\right),\ \forall j \in \Omega_{\mathrm{RE}}$	(6)
	$k_{\mathrm{b},j} = \frac{k_{\mathrm{f},j}}{K_j}$ and $K_j = \exp\left(-\frac{\Delta_{\mathrm{r}} G_j(T)}{\mathrm{R}T}\right),\ \forall j \in \{\mathrm{CSD,\ PFD}\}$	(7) and (8)
	$\Delta_{\mathrm{r}} G_j(T) = \Delta_{\mathrm{r}} H_j(T) - T\Delta_{\mathrm{r}} S_j(T),\ \forall j \in \{\mathrm{CSD, PFD}\}$	(9)
Energy balance	$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\sum_{j\in\Omega_{\mathrm{RE}}} q_j}{C_{\mathrm{p,bat}}}$	(10)

ref. 43, and we set the threshold for the battery self-heating rate to 0.02 K min^{-1}. Furthermore, adiabatic conditions are assumed for the self-heating phase. From this, the energy balance is computed based on eqn (10). For the sake of brevity, the following dependencies are only briefly summarised. The corresponding model equations can be found in the ESI.† Temperature changes during self-heating and thermal runaway are driven by the heat of the single reactions. According to Hess' law, molar heat of reaction is computed from the enthalpies of formation of participating species. The heat consumption or production rate originating from each reaction is then calculated as the product of the molar heat of the reaction and reaction rate. The heat capacity of the battery is calculated from the individual components. For the calculation of concentrations and partial pressures and the distinction between their influence on reactions, four reference phases are assumed: the anode including the SEI, the liquid electrolyte, the cathode, and a gas phase. Here, the volume refers to the volume of the anode + SEI, electrolyte or cathode, depending on the species for which the concentration is calculated. Gas/liquid phase equilibrium is assumed. Eventually, the solubility of each gaseous species in the electrolyte is calculated assuming they stay in the liquid phase until the maximal solubility is reached. Further model assumptions are summarised in the following:

- All reactants are ideally mixed, so no spatial discretisation is applied. Since the time constant for diffusion and thermal conductivity is on the order of seconds and the ARC measurement takes hours, we deem the assumption of a perfectly mixed system a sound approach.
- A pressure increase due to gas evolution does not impact liquid-phase reactions.
- All reactions except for $LiPF_6$ and PF_5 decomposition are irreversible.
- SEI formation reactions are modelled as chemical reactions.

2.3 Experiments and parameterisation

This study aims to reveal the influence of H_2O impurities and SEI composition on battery safety at high temperatures. Kinetic parameters, k_0 and E_A, are obtained from two distinct experimental studies: Stich *et al.*[31] focused on the decomposition of $LiPF_6$ with H_2O as an impurity, providing an isolated study of $LiPF_6$, PF_5 and POF_3 decomposition. Meanwhile, Maleki *et al.*[33] performed an ARC measurement encompassing all concurrent reactions. In the following, the experiments themselves are described, as well as our parameterisation procedure.

Stich *et al.*[31] used a 1 : 1 v:v EC:DEC 1 M $LiPF_6$ electrolyte with an initial concentration of ≤ 15 ppm water and ≤ 50 ppm HF. They then added 1000 ppm of water. The concentrations of H_2O, HF and HPO_2F_2 have been monitored for 15 days at room temperature by coulometric Karl Fischer titration, acid–base titration, and ion chromatography.

An ARC measurement by Maleki *et al.*[33] is used to parameterise and study the transition from a safe battery at room temperature over its self-heating phase starting around 100 °C until the transition into rapid thermal runaway. The ARC measurements are conducted by inserting the battery into an oven-like apparatus. Following this, a linear temperature gradient is usually applied to reach a temperature where the first degradation reactions are expected to occur. In the experiments of Maleki *et al.*,[33] this was 40 °C. Then, a heat–wait–seek (HWS) procedure is applied, where a 10 °C step-wise increase in temperature is conducted. Each step is followed by a period where the battery cell is equilibrating with the new temperature, and the temperature is monitored. No further heating is supplied when self-heating is detected during the seeking

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procedure. A detailed explanation of the underlying algorithm and the experiment used to parameterise our model is given in our previous publication.[4] The cell used by Maleki *et al.*[33] had a capacity of 550 mA h. The chemistry was reported to be a carbon-based anode, a $LiCoO_2$ cathode, and polyvinylidene fluoride as the binder in both electrodes, and an ethylene carbonate (EC)/ethyl methyl carbonate (EMC) mixture with $LiPF_6$ as the conductive salt. Since the original publication lacked some data crucial for simulation, the following assumptions have been made:

• The electrolyte consists of a 50 : 50 EC : EMC (v:v) mixture with 1200 mol m^{-3} $LiPF_6$.

• The geometry of the cell is taken from a commercially available one that has a capacity and cell chemistry comparable to the cell used in the study by Maleki *et al.*[44]

• The electrode geometry is similar to other published electrodes of the same chemistry[45,46]

The kinetic parameters for the decomposition of $LiPF_6$ and the subsequent reactions of PF_5 and POF_3, namely CSD, PFD, and POFD, have been parameterised such that two constraints are met. The first is that the experimental data[31] of changes in the electrolyte at room temperature could be reproduced. The second is that the reactions notably accelerate within the reported temperature interval (see Table 1). For the decomposition of Li_2CO_3, the kinetic constants are set such that this reaction is not limiting.

For the other reactions, the kinetic parameters have been adjusted to meet two constraints: the first one is that the reactions occur within the reported temperature interval for each given reaction (Table 1). The second constraint is that the simulation can reproduce the experimental data from Maleki *et al.*[33] Exceptions to this are the SEI forming reactions, OSP, ISP, and LSP. Given that they are, in nature, electrochemical reactions, they should notably occur even at room temperature with an unprotected electrode. Reported energy barriers for these reactions are very low, if not 0.[17,18,47] The term introduced to account for the inhibition effect of the SEI (see eqn (3)) does not prevent notable reactions even at low temperatures with these low barriers. Therefore, the energy barriers are adjusted such that these reactions occur in a temperature range concurrently with the decomposition of the existing SEI.

The identified parameter values are given in the ESI.† Model equations and parameters were implemented in MATLAB, and the simulation was performed using the ode15s solver. All calculations have been performed with MATLAB Version 2022a,[48] using an i7-9750H processor with 16 GB RAM. The average simulation time was six minutes.

2.4 Reference conditions and variation in water content and SEI properties

With their intricate internal decomposition mechanisms during formation, operation and thermal abuse, Li-ion batteries present a complexity that hinders the precise determination of the amount of species present at the beginning of a thermal decomposition study. Due to these restrictions, educated estimates for some components are necessary. For the sake of brevity, the procedures to arrive at the initial conditions will be shown in the ESI.† The values of interest are the conductive salt decomposition products, namely PF_5, POF_3, HPO_2F_2, HF, and H_2O and the initial amount of the SEI components, Li_2O, LEDC, LiOH, LiF, and Li_2CO_3, as well as the initial SEI thickness, see Table 3.

H_2O impurities, SEI compositions and SEI thickness can be influenced during manufacturing processes such as electrode drying or formation, as well as through battery ageing. At present, it is unfeasible to model and incorporate all these processes. Nevertheless, we implicitly investigate the effects and variability of different manufacturing conditions on battery behaviour during thermal abuse by designing a case study that uses various combinations of manufacturing and ageing inspired initial values for water impurity, SEI composition and thickness (Table 3). The variation in H_2O content is based on a study of different drying procedures by Huttner *et al.*[32] The H_2O content will change due to different water contents in the electrodes following Huttner's results for the undried "wet", the medium dried, and the highly dried electrodes. The different values in the anode of 2422 ppm, 286 ppm, and 214 ppm translate into 5 vol%, 0.6 vol% and 0.4 vol% of LiOH content within the SEI. The reported values of 2644 ppm, 500 ppm and 464 ppm in the separator and 313 ppm, 156 ppm, and 63 ppm for the cathode translate into an H_2O increment in the electrolyte of 930 ppm, 334 ppm, and 172 ppm, respectively.

In the early 2000s it was revealed that LEDC is the primary decomposition product of EC.[49] A recent study by Wang *et al.*,[50] however, questions the stability of LEDC and proposed that lithium ethylene mono-carbonate (LEMC) is the stable alternative. Following this, Xie *et al.*[51] investigated the formation pathways of LEMC, including the decomposition of LEDC and found that all kinetically favourable pathways need water as a reactant. It is apparent that there is still a lively discussion on the exact composition of the SEI. Thus, we will consider a wide range of differing compositions and assume LEDC as the major organic compound: here, 90 vol%, 45 vol% and 0 vol% LEDC are investigated. We did not choose the pure LEDC content since LiOH has to be part of the SEI to account for water impurities. Furthermore, the complete lack of Li_2CO_3 would lead to an accumulation of HF according to the above-described behaviour, which we deem very improbable. Thus, the remaining volume percentages are Li_2CO_3.

The SEI thickness has been varied between 25 nm, 50 nm, and 75 nm, which are within reported values for SEI thicknesses.[10,52,53]

All variations undergo the same formation and conditioning procedure described in the ESI.† From this, the initial values listed in Table 3 are calculated.

An extensive parameter variation, including all possible 27 variations (3 × 3 × 3), has been simulated to investigate the interdependence. The additional 18 initial values, apart from the 9 already listed in Table 3, can be found in the ESI.†

The initial and maximum temperatures are 25 °C and 220 °C, respectively. The pressure is constant at $p = 101\,325$ Pa.

The initial conditions for all modelled species and their physical parameters and data on the electrode structure can be found in the ESI.†

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